

DataTalk Evaluation Metrics Overview

DataTalk was evaluated on a benchmark of 45 natural-language questions testing different aspects of sql like aggregation , joining , filtering , basics ..

By this dataset → i want to check whether the system can produce a query that executes & returns expected results

Results

Metric	Result	Meaning
Total questions	45	-----
Text-to-SQL accuracy	8.89%	Exact match with the reference SQL
Execution accuracy	53.33%	Generated query returns the expected result
First-pass accuracy	53.33%	Questions answered correctly without a retry
Retry success rate	0	Failed queries recovered successfully through retry
Average retry count	0	Average number of retry attempts per question

Interpretation

1. Text-to-SQL Accuracy : 8.89%

A query can be logically valid but still fail exact matching because of an alias, column selection, JOIN formulation, ordering, or another difference in SQL representation.
so like two outputs can represent the same intended calculation, but they are not textually identical

2. Execution Accuracy : 53.33%

more than half of the evaluated questions reached the expected result on the first evaluation path. The gap between 8.89% exact SQL accuracy and 53.33% execution accuracy shows that the system often produces SQL that differs from the reference

formulation while still being able to answer correctly

3. First-Pass Accuracy : 53.33%

In this evaluation, first-pass accuracy is equal to execution accuracy at 53.33%. This means the successful cases were completed directly, while the retry mechanism was not contributing to the measured accuracy for this particular dataset .

4. Retry Success Rate + Average Retry Count

No successful retry recoveries were recorded during the 45-question evaluation run because the benchmark did not trigger any retry attempts (`average_retry_count = 0`). However, this does not mean that the SQL retry and embedding-based memory system is not working.

The retry mechanism was tested separately using Pytest, where the SQL memory successfully retrieved a relevant previous example from the embedding-based vector store. This confirms that the embedding retrieval component and SQL memory retrieval are functioning correctly.

→Therefore:

0% retry success rate reflects the behavior of this particular evaluation dataset, not a failure of the retry or embedding-retrieval implementation.

Prepared By :

Manar ElFAKih Romdhane